

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (Civil-Structure) Examination

January 2014

CL704 Advanced Concrete Technology

Date: 06.01.2014, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q – 1 Enlist advantages and disadvantages of concrete use. Explain grade of concrete. [07]
- Q - 2 (a) Explain hydration of cement incorporating rate of hydration and mechanism of hydration. [07]
- Q - 2 (b) Attempt any One [07]
- (i) Enlist six special concrete types and explain any one in detail.
 - (ii) Enlist ingredients of modern concrete. Explain role of each ingredients.
- Q - 3 Attempt any Two [14]
- (i) Define Chemical Admixture. Explain its advantages during hardened and fresh state of concrete.
 - (ii) Explain Marsh Cone Test to decide Super plasticizer Saturation Point.
 - (iii) Explain Effects of GGBS on fresh and Hardened Concrete.

SECTION – II

- Q – 4 Write a note on viscosity modifying agents and corrosion inhibitors. [07]
- Q - 5 (a) Explain rebound hammer test including use, procedure of test, result interpretation from reading and limitation of test. [07]
- Q - 5 (b) Attempt any One [07]
- (i) Design a concrete mix as per IS:10262-2009, I.S. method with following data:
Characteristic Strength at 28 Days = 35 MPa
Maximum size of Aggregate = 20mm, Degree of workability = 100mm slump, Quality Control/Supervision = Very Good, Type of Exposure = Severe, Method of Placing = Manual
Type of Aggregate = Crushed Angular, Maximum Cement Content = 450 Kg/m³, Chemical Admixture Type = Super plasticizer, 1% by mass of cementitious material, Specific Gravity of Cement = 3.15, Course Aggregate = 2.80, Fine Aggregate = 2.60, Super plasticizer = 1.15 & sand is of grade zone III.
 - (ii) Compare between IS, ACI and DOE method for concrete mix design
- Q - 6 Attempt any Two [14]
- (i) Explain the compression test on cube and cylinder and discuss the possible reasons for different strength in case of test on cube and cylinder from the same concrete.
 - (ii) Enlist factors affecting durability of concrete and explain any one in detail.
 - (iii) Explain carbonation effect on corrosion and how it is detected using phenolphthalein test.
